Seva First Innovation Challange 2026

Nir-ikshan

Inspect. Act. Verify. Restore.

1 Theme Swachh & Sustainable Bharat (Clean and Sustainable Bharat)

(Track A)

Team : AURAlytics | Dept. of CS

Nir-ikshan

The Problem

- Recharge assets are built and waterbodies are restored, but their condition is **not consistently tracked**
- **Maintenance ownership is unclear** once the work is handed over
- **Recurring waste and sewage inflow** continue after restoration
- **Work is recorded as complete without checking** the actual outcome
- **Records sit in different offices**, with no single accountable workflow

Kammasandra Lake

polluted bank of the Yamuna river with toxic foam floating on it

Nir-ikshan

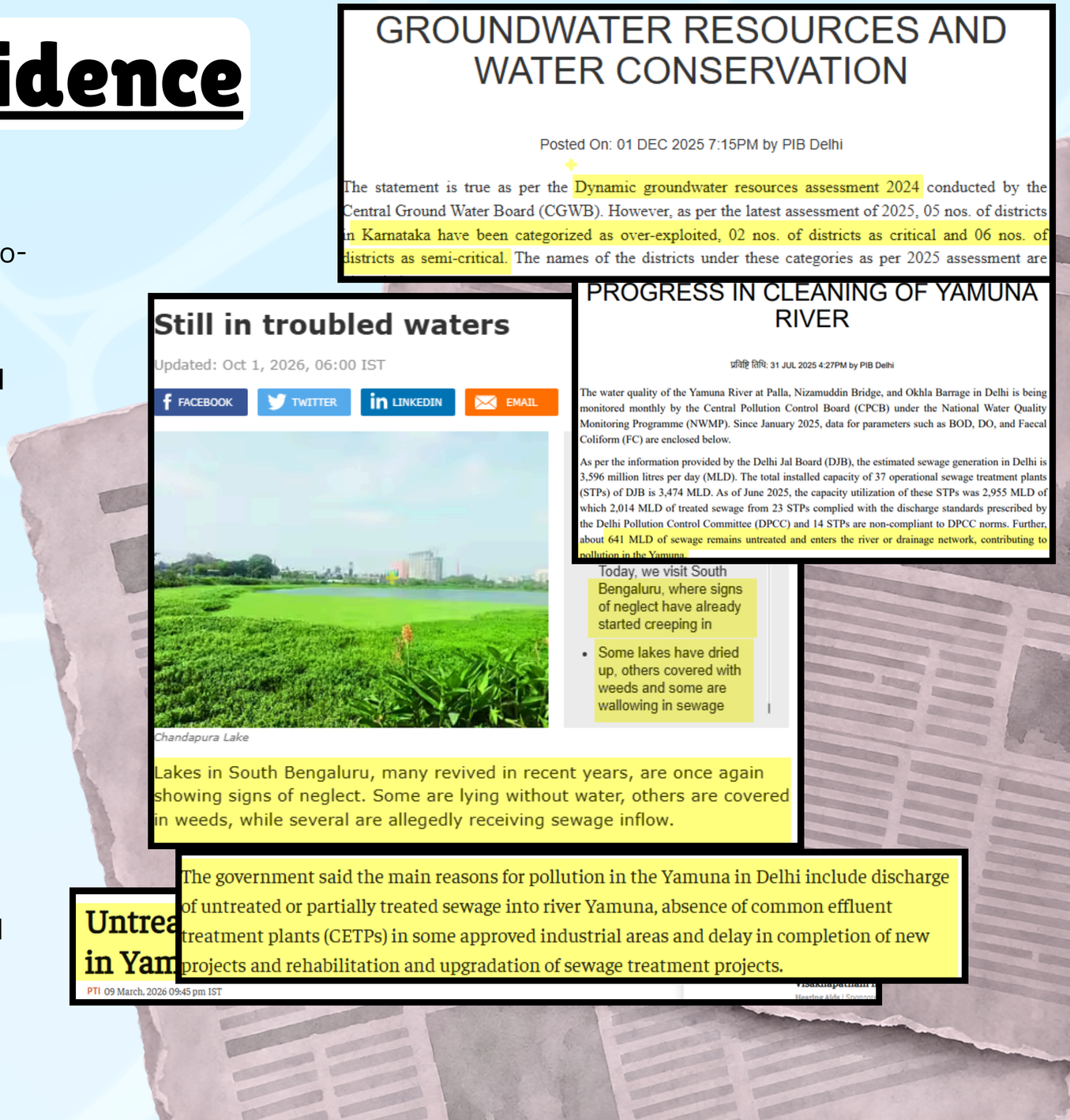
Evidence

- The 2025 assessment classifies **five Karnataka districts as over-exploited:** Kolar, Bengaluru Urban, Chikkaballapura, Bengaluru Rural and Chitradurga. This gives a solid basis for a targeted micro-watershed pilot, not a generic awareness campaign.

- News reporting from **Bengaluru** documents lakes **facing renewed neglect despite rejuvenation efforts, with alleged untreated sewage discharge and citizen petitions** to the NGT. Nir-ikshan would create a post restoration source-control and verification ledger so such recurrence becomes visible and actionable

- **Even in a heavily monitored river system**, government reporting identified about **641 MLD of untreated sewage entering the Yamuna or drainage network.** Nir-ikshan helps local bodies track such source-control gaps and verify corrective action.

- Reporting stated that **untreated or partially treated sewage containing detergents** is a key reason for recurring toxic froth in the Yamuna. **The government also cited delayed completion and rehabilitation/upgradation of sewage-treatment projects as contributing factors.**



Nir-ikshan

Proposed Solution

What Nir-ikshan is ?

- A GIS-enabled workflow for inspection, action and verification of water assets
- Turns asset and restoration records into accountable, measurable public outcomes

What it does

- Helps local authorities identify and verify whether priority recharge assets are functional and maintained
- Focuses on areas where district-level classification still shows stress

What it does not do

- Does not replace groundwater regulation
- Puts existing recharge and rainwater-harvesting responsibilities into a repeatable

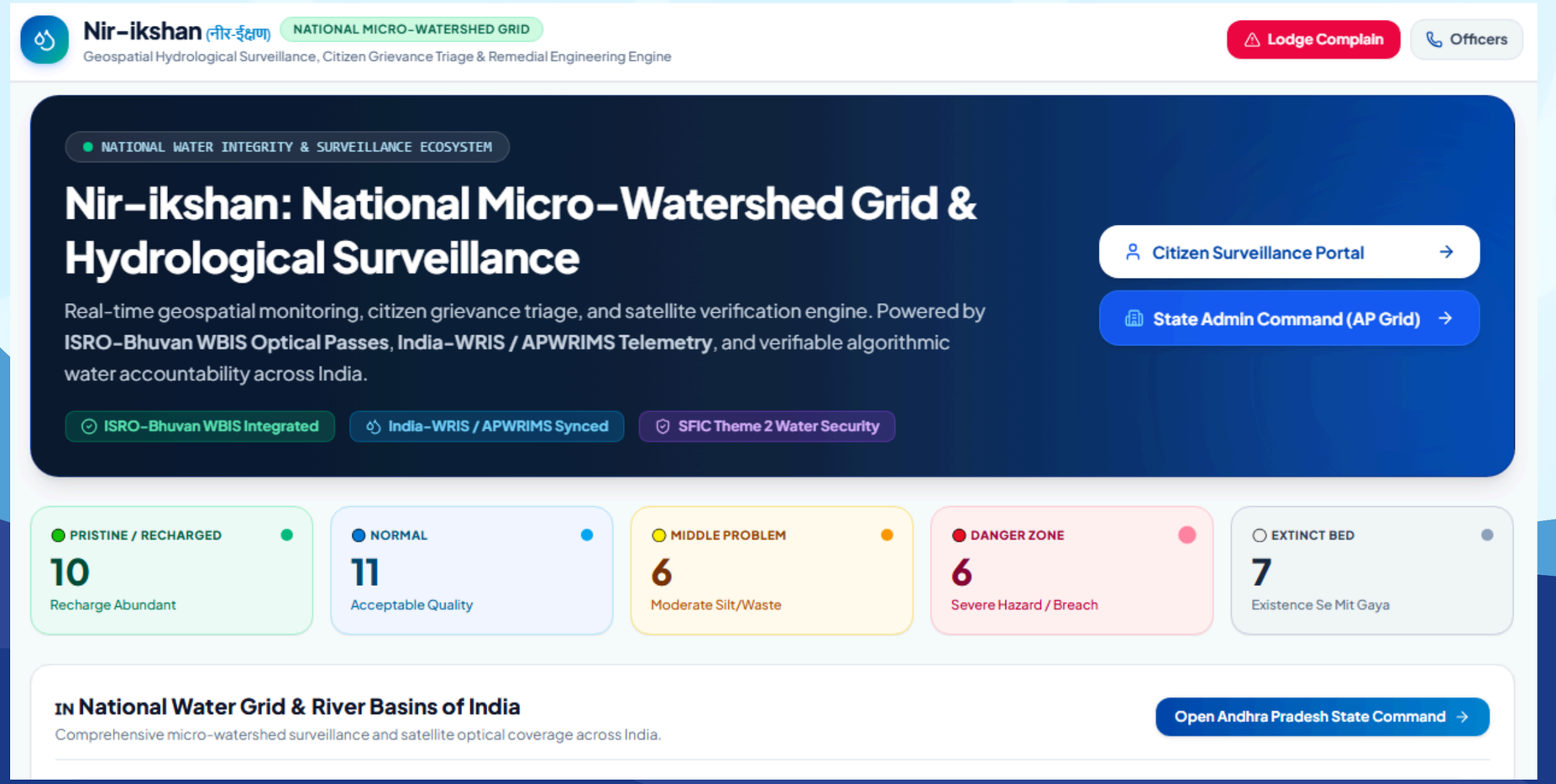
inspect → act → verify cycle

Nir-ikshan

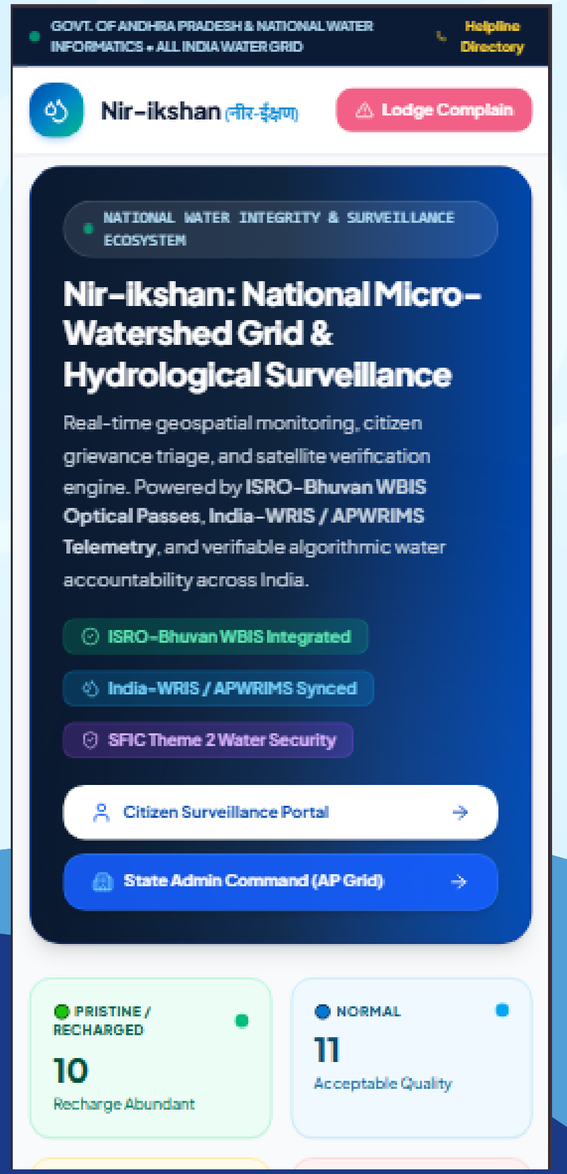
Product Interface

Link : https://nir-ikshan.vercel.app/

Desktop View



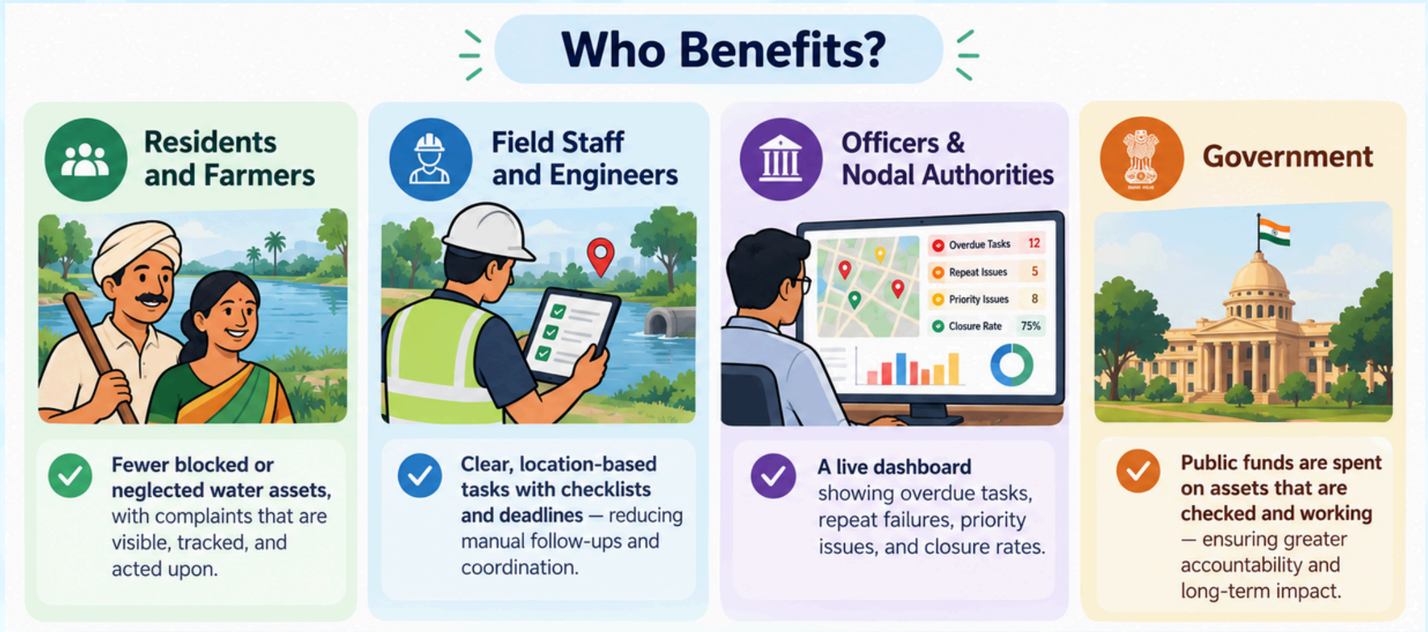
Smart Phone View



Impact & Benefits

What changes when Nir-ikshan is implemented

Today	With Nir-ikshan
Assets are scattered across files and offices	One map-based register of every asset
Nobody owns the fix	Every issue has a named owner and deadline
Work is closed on paper	Closure needs photo proof and verification
Public can't see progress	Public can see status and raise complaints
Repeat failures go unnoticed	Repeat issues are flagged and escalated



Long-term and indirect benefits (expected)

- **Local jobs and livelihoods:** regular inspection, cleaning and repair create steady work for field inspectors, maintenance workers and local contractors, and give community volunteers a role
- **Cleaner waterbodies**: tracking waste and inflow sources and fixing them helps improve conditions, which supports better water quality
- **Less pressure on groundwater**: well-maintained recharge assets and restored waterbodies can supply surface water and recharge, so communities depend less on borewells

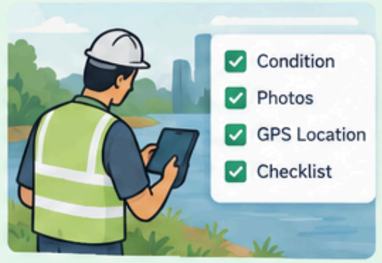
Nir-ikshan

Workflow

1 Identify
(Public / Inspector)

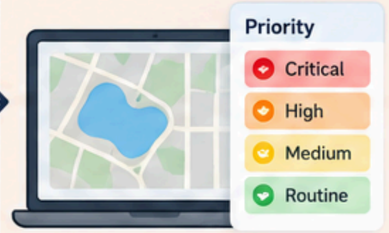
- Citizen files a complaint or inspector finds a problem during routine check.
- Asset is located on the map (waterbody / drain / well etc.).

2 Inspect
(Field Inspector)



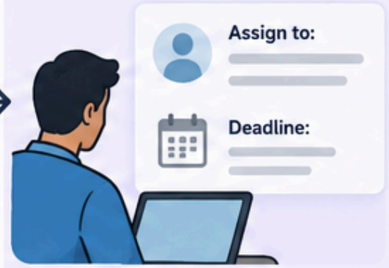
- Visits the site.
- Records condition, photo and GPS using a checklist.

3 Prioritise
(System)



- System assigns a priority using transparent rules (based on condition, impact, risk, usage etc.).

4 Assign
(Nodal Officer)



- Gives the issue a named owner (department / worker) and a deadline.

5 Act
(Action Worker)

- Does the rspair or cleaning.
- Uploads before/after proof (photos, notes).

6 Verify
(Field Inspector)

- Checks the work on site.
- Accepts it, or rejects it if not satisfactory.

Rejected (sends back to Step 5)

Accepted

7 Review
(Admin / Nodal Officer)

- Dashboard shows overdue tasks, repeat failures and closure rates.
- Feeds into seasonal review and helps plan future work.

Continuous Monitoring & New Reports

Nir-ikshan

Technologies & Data Used

Users

(5 Roles)

- **Admin**
- **Nodal Officer**
- **Inspector/ Verifier**
- **Action Worker**
- **Public User**

(Frontend)

- **React 19** and TypeScript, built with Vite
- **Tailwind CSS** v4, responsive layout with dark and light themes
- **Lucide icons** and Motion for animation

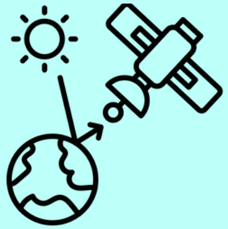
(Backend)

- **Node.js 22** with **Express**, written in **TypeScript**
- **REST API** with proxy routes for water-data sources (India-WRIS and ISRO Bhuvan)
- **Colour Coded Status**

(API)

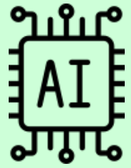
- **India-WRIS**: reservoirs, river discharge, groundwater depth, rainfall
- **ISRO Bhuvan:** satellite passes and the NDWI water index

(GIS & Mapping)



- Custom SVG **vector map covering 54 mandals across 3 districts** (Parvathipuram, Manyam, Vizianagaram, Visakhapatnam)
- Rivers, reservoirs and coastline layers, with zoom and pan controls
- **5-colour status classification for each water body**

(AI)



- **Google Gen AI SDK** for NDWI-based water-spread classification and complaint triage, with human review

Nir-ikshan

Refrences

- https://sansad.in/getFile/lsapps/loksabhaquestions/annex/187/AS542_ZqoVDN.pdf?source=lsapps
- Untreated sewage, detergents behind recurring toxic froth in Yamuna: Govt tells RS – ThePrint – PTIFeed
- https://www.pib.gov.in/PressReleaseDetail.aspx?PRID=2197180®=6&lang=1
- https://www.pib.gov.in/PressReleasePage.aspx?PRID=2150744
- https://theprint.in/india/untreated-sewage-detergents-behind-recurring-toxic-froth-in-yamuna-govt-tells-rs/2874343/
- https://bangaloremirror.indiatimes.com/bangalore/cover-story/still-in-troubled-waters/articleshow/134601463.cms
- https://www.pib.gov.in/PressReleasePage.aspx?PRID=2150744®=48&lang=2
- https://www.indiatoday.in/india/story/polluted-river-stretches-india-fall-351-296-rajya-sabha-ptag-2952176-2026-07-20
- https://theprint.in/india/untreated-sewage-detergents-behind-recurring-toxic-froth-in-yamuna-govt-tells-rs/2874343/
- https://bhuvan-app1.nrsc.gov.in/api/#
- https://indiawris.gov.in/wris/#/home